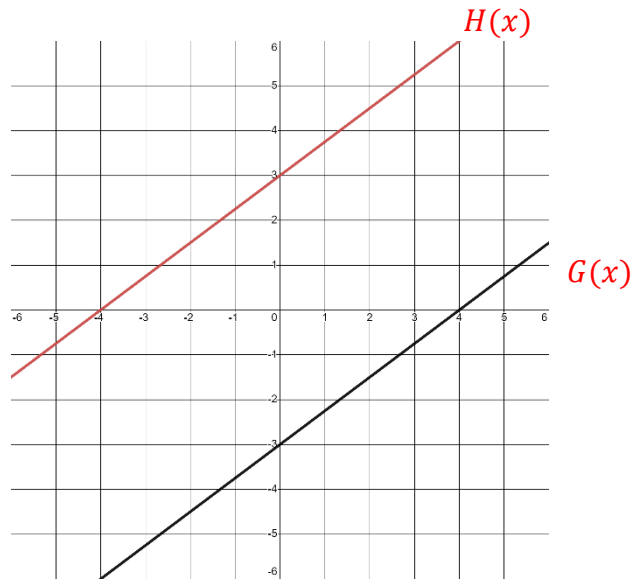


Algebra 1  
Unit 13  
Lesson H1 Practice

Name Answer Key  
Date \_\_\_\_\_  
Period \_\_\_\_\_

The graph of the function  $G(x)$  is shown.



1. Write the function,  $G(x)$ , that represents the line.

$$G(x) = \frac{3}{4}x - 3$$

2. Draw a line that is parallel to  $G(x)$  with a y-intercept of 3. Label this line  $H(x)$ .  
3. Describe how  $G(x)$  can be shifted to coincide with  $H(x)$ .

Shift up 6 units

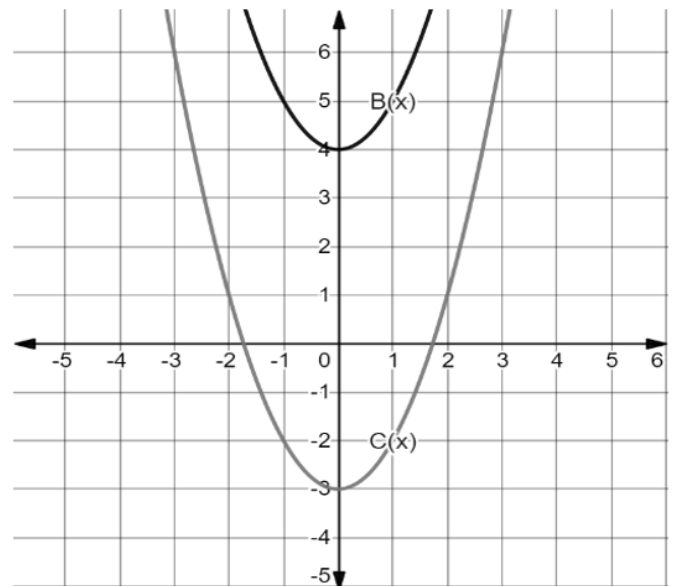
4. Write the function  $H(x)$  in terms of  $G(x)$ .

$$H(x) = G(x) + 6$$

5. The graphs of  $B(x)$  and  $C(x)$  are shown on the graph.

Write the transformation by completing the equation.

$$C(x) = B(x) + (-7)$$



The functions  $m(x)$  and  $p(x)$  are shown on the graph.

6. Write  $m(x)$  in vertex form.

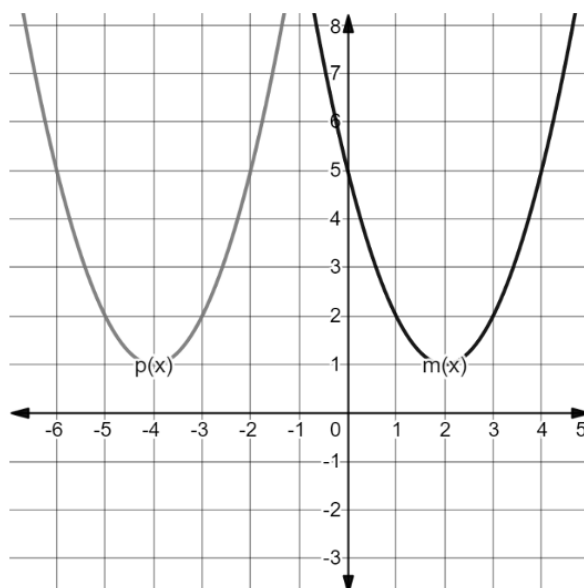
$$m(x) = (x - 2)^2 + 1$$

7. Write  $p(x)$  in vertex form.

$$p(x) = (x + 4)^2 + 1$$

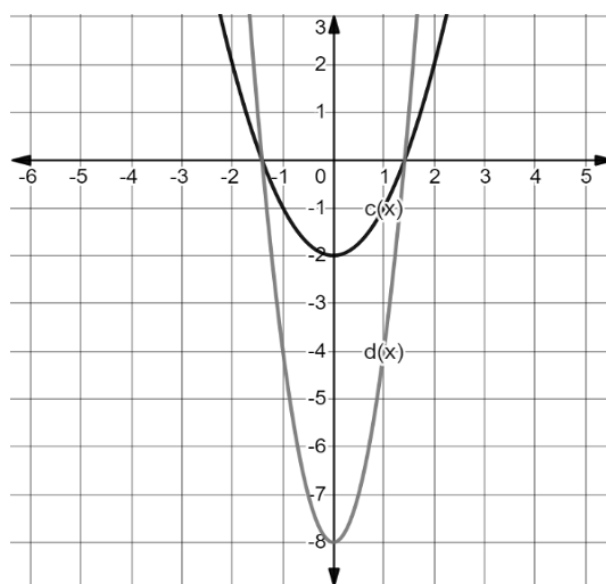
8. Describe how  $m(x)$  was transformed to create  $p(x)$ .

Shifted left 6 units



9. The functions  $c(x)$  and  $d(x)$  are shown on the graph, where  $d(x)$  is a transformation of  $c(x)$ . What is the value of  $k$  such that  $d(x) = kc(x)$ ?

$$k = 4$$



10. The table of values for  $d(x)$  and  $f(x)$  are shown, where  $f(x)$  is a transformation of  $d(x)$ . Write a description of the transformation.

| $y = d(x)$ |     | $y = f(x)$ |     |
|------------|-----|------------|-----|
| $x$        | $y$ | $x$        | $y$ |
| -2         | 16  | -2         | 20  |
| -1         | 4   | -1         | 8   |
| 0          | 0   | 0          | 4   |
| 1          | 4   | 1          | 8   |
| 2          | 16  | 2          | 20  |

$d(x)$  is shifted up 4 units to form  $f(x)$ .